CAN WEARABLE TECH ACTUALLY PREVENT INJURIES

More tracking data doesn't always mean fewer injuries. Here's what the research tells us.



WHAT DO WEARABLES ACTUALLY TRACK

Different wearable sensors collect different types of movement and training data



DISTANCE

How far an athlete travels during training or competition

SPEED

How fast an athlete runs, including high-speed efforts.

ACCELERATION

How quickly an athlete speeds up or slows down.

JOINT MOTION

Estimates of joint angles and movement patterns.

Tracking movement isn't the same as predicting injury.

TRACKING IS NOT PREVENTION

Collecting data, identifying risk, and reducing injuries require different kinds of evidence.

01 - TRACK

Measure movement and training workload.

What happened?

02 - PREDICT

Use patterns to estimate potential injury risk.

What might happen?

03 - PREVENT?

Demonstrate that data-guided decisions actually reduce injuries.

Did the intervention work?

An accurate measurement is not proof of injury prevention

HOW ACCURATE ARE WEARABLE SENSORS?

Accuracy can change depending on the movement, joint, and sensor calibration.

ONE STUDY FOUND MEASUREMENT DIFFERENCES RANGING FROM

<0.1° to 24.1°

Difference between wearable IMU and laboratory motion-capture joint-angle measurements

Ekdahl et al. (2023) · 11 participants

WHY DO READINGS DIFFER?

CALIBRATION

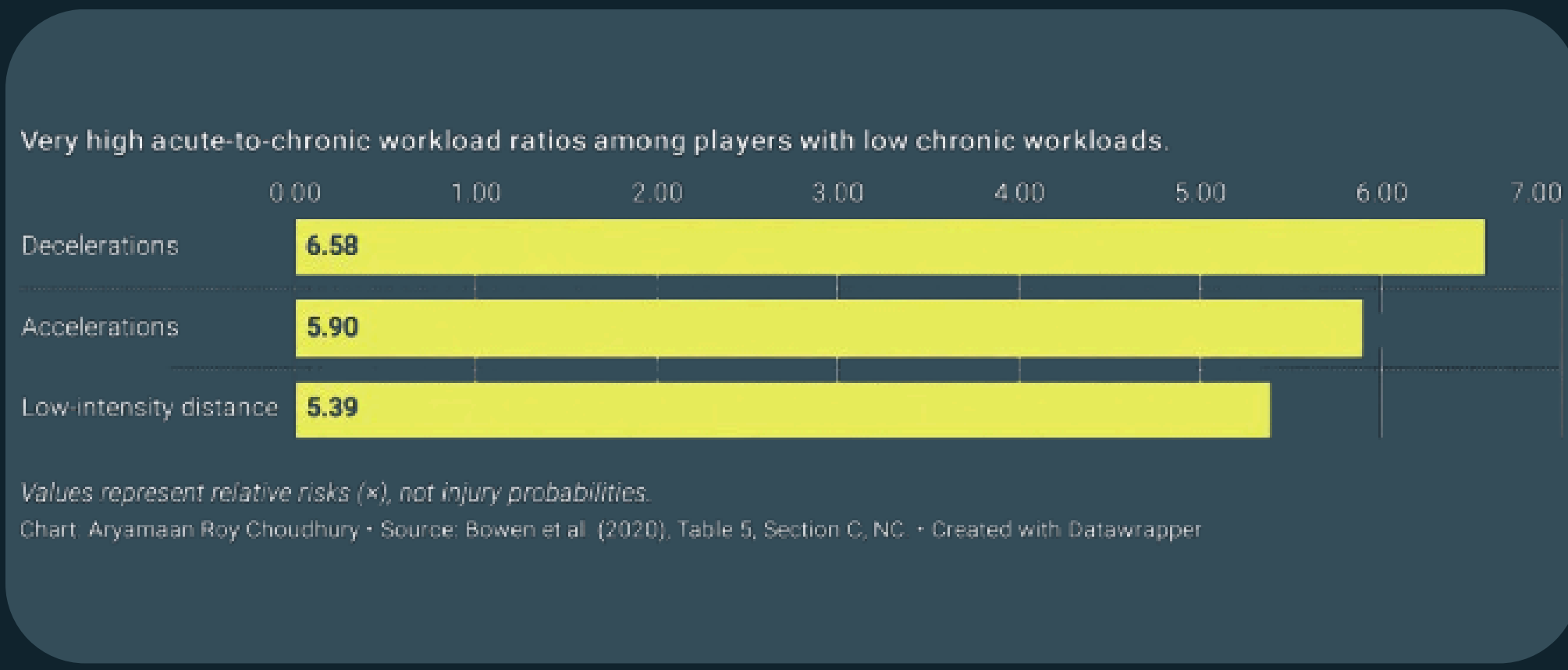
MOVEMENT

JOINT / ANGLE

A sensor that works well for one movement may be less accurate for another.

WHEN WORKLOAD SPIKES, INJURY RISK CAN RISE

Sudden increases in training workload may be associated with higher injury risk.



5-7x

Higher relative non-contact injury risks observed under these specific workload conditions

33 professional footballers · 3 seasons

Higher observed injury risk does not mean that wearable monitoring has been proven to prevent injuries.

WHAT WEARABLE DATA CAN AND CAN'T TELL US

Wearables provide useful information, but the data has important limitations.

CAN HELP

TRACK WORKLOAD
Monitor how much an athlete trains.

SPOT CHANGES
Identify unusual workload patterns.

INFORM DECISIONS
Provide context for training adjustments.

CAN'T GUARANTEE

EXACT INJURY RISK
Readings don't predict every injury.

UNIVERSAL ACCURACY
Measurements vary by task and sensor setup.

INJURY PREVENTION
Tracking alone doesn't prove that injuries will be prevented.

Use wearable data as a tool for decision-making - not as a guarantee of athlete safety.

USE THE DATA. DON'T BLINDLY TRUST IT.

Wearable technology works best as a tool to support informed coaching decisions.

01 - CHECK THE MEASUREMENTS



Use sensors validated for the movement and metric being tracked.

02 - LOOK AT THE PATTERN



Compare recent workloads with the athlete's training history.

03 - ADD HUMAN JUDGMENT



Consider athlete feedback and professional assessment before making changes.

BETTER DATA + BETTER JUDGMENT

More informed decisions - not guaranteed injury prevention.

THE RESEARCH BEHIND THE DATA

Explore the studies behind the findings in this carousel.

FEATURED RESEARCH

Ekdahl et al. (2023)
How sensor calibration affects movement measurement accuracy.
Panel 4 · Sensor accuracy

Bowen et al. (2020)
How training workload patterns relate to injury rates.
Panel 5 · Workload and injury risk

ADDITIONAL STUDIES

Chia et al. (2021)
Knee and trunk measurement validity

Macadam et al. (2019)
Sprint sensor reliability

Mason et al. (2024)
Running gait measurements

Kim et al. (2026)
IMU-based force estimation

SCAN FOR THE FULL REFERENCE LIST
Research links, study details, and further reading